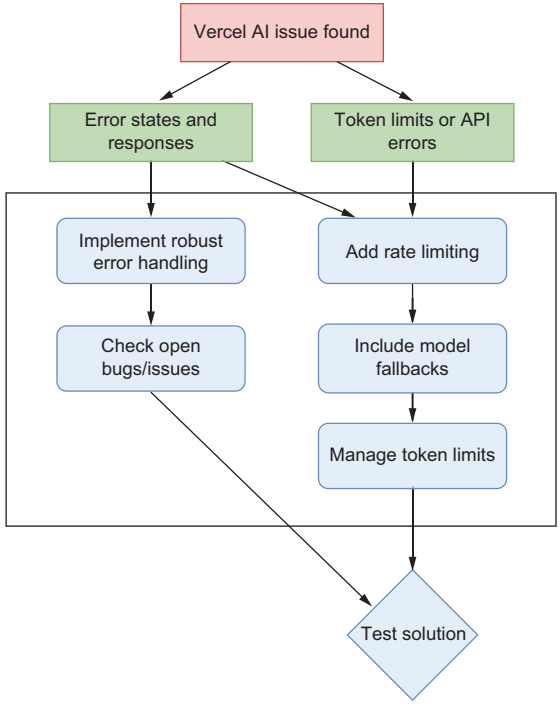


Figure 8.5 Troubleshooting strategies for Vercel AI SDK problems. There are two main paths: error states/responses and token limits/API errors. Each path shows specific steps for resolution, including implementing error handling, rate limiting, model fallbacks, and token management.

You can use this figure as a starting point when debugging problems with Vercel AI SDK implementation and relevant backend services. Let's explore the two main paths: errors states and responses, and token limits and API errors.

8.2.1 Handling error states in AI-generated content

When working with AI models through the Vercel AI SDK, the most frequent problems come from the misalignment between what providers advertise and what they support. Their API can change, but they do not inform the endpoint consumers about the implications of those changes. Of course, this challenge might not exist at all if you self-host your own LLM, but it can appear when there are intermediate errors that break the normal flows.

PROVIDER COMPATIBILITY PROBLEMS

The following is a comprehensive breakdown of common error states you may find when working with AI providers. The most prevalent source of errors occurs when AI providers make unannounced changes or have undocumented limitations:

- Sudden API changes without warning (especially common with OpenAI and Anthropic)
- Regional restrictions that aren't clearly documented (e.g., Azure OpenAI Services not being available in certain regions)